

Please write clearly in block capitals.

Centre number

--	--	--	--	--

Candidate number

--	--	--	--

Surname

Forename(s)

Candidate signature

I declare this is my own work.

INTERNATIONAL GCSE CHEMISTRY

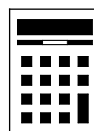
Paper 1

Thursday 3 November 2022 07:00 GMT Time allowed: 1 hour 30 minutes

Materials

For this paper you must have:

- a pencil and a ruler
- a scientific calculator
- the periodic table (enclosed).



Instructions

- Use black ink or black ball-point pen.
- Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- If you need extra space for your answer(s), use the lined pages at the end of this book. Write the question number against your answer(s).
- Do all rough work in this book. Cross through any work you do not want to be marked.
- Show all your working.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 90.
- You are expected to use a scientific calculator where appropriate.
- A periodic table is provided as a loose insert.

For Examiner's Use

Question	Mark
1	
2	
3	
4	
5	
6	
7	
TOTAL	



N 0 V 2 2 9 2 0 2 1 0 1

Answer **all** questions in the spaces provided.

Do not write
outside the
box

0 1

This question is about pure substances and mixtures.

0 1 . 1

Which substance is a mixture?

[1 mark]

Tick (✓) **one** box.

Air

☐

Carbon dioxide

☐

Sodium chloride

☐

Water

☐

0 1 . 2

Which method can be used to separate a mixture of sand and water?

[1 mark]

Tick (✓) **one** box.

Crystallisation

☐

Filtration

☐

Titration

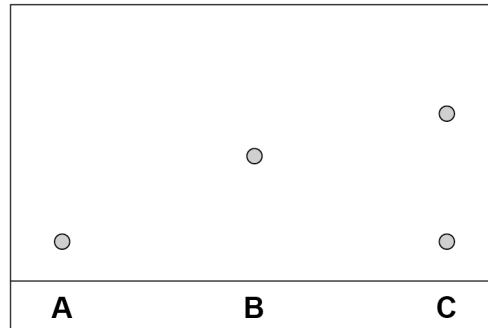
☐

0 1 3

A student used chromatography to investigate three food colourings, **A**, **B**, and **C**.

Figure 1 shows the results.

Figure 1



Give **three** conclusions about food colouring **C**.

[3 marks]

- 1 _____
- 2 _____
- 3 _____

Question 1 continues on the next page

Turn over ►



0 1 . 4

A different student measured the melting point of four solid food colourings, **D**, **E**, **F**, and **G**.

Table 1 shows the results.

Table 1

Food Colouring	Melting Point °C
D	80
E	84–87
F	139
G	72–76

Which food colourings are pure?

Give **one** reason for your answer.

[2 marks]

Pure food colourings _____

Reason _____



0	1	5
---	---	---

A medicine contains 4.2% by mass of a food colouring.

The medicine has a mass of 1.2 g.

Calculate the mass of food colouring in the medicine.

Give your answer in mg.

1000 mg = 1 g

[3 marks]

Mass of food colouring = _____ mg

10

Turn over for the next question

Turn over ►

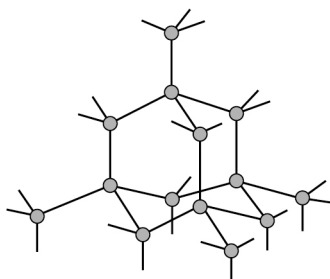


0 2

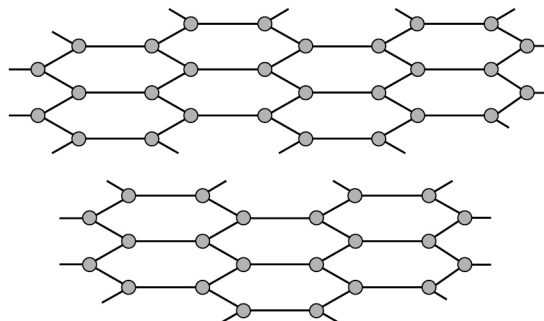
Diamond and graphite have giant covalent structures made of carbon atoms.

Figure 2 represents the structures of diamond and graphite.

Figure 2



Diamond



Graphite

0 2 . 1

Give **two** differences between the **structures** of diamond and graphite.

Use **Figure 2**.

[2 marks]

- 1 _____
- 2 _____

0 2 . 2

Which is a property of diamond?

[1 mark]

Tick (✓) **one** box.

Conducts electricity

☐

Dissolves in water

☐

Hard

☐


0 2 . 3 Explain why diamond has a high melting point.

Use **Figure 2**.

[3 marks]

0 2 . 4 Graphite also has a high melting point.

Give **one** other property of graphite.

[1 mark]

0 2 . 5 Which of the following has a giant covalent structure?

[1 mark]

Tick (✓) **one** box.

Chlorine

☐

Iron

☐

Magnesium oxide

☐

Silicon dioxide

☐

Turn over for the next question

Turn over ►



There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

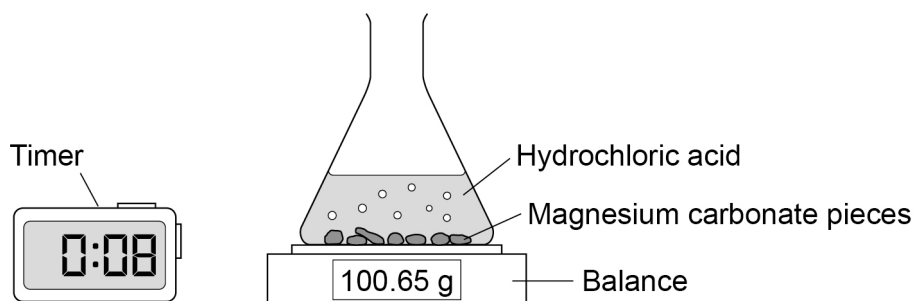


0 3

A student reacted magnesium carbonate with hydrochloric acid.

Figure 3 shows the apparatus.

Figure 3



0 3 . 1

Suggest **one** safety precaution the student should take when doing the experiment.

[1 mark]

0 3 . 2

Name the gas produced when magnesium carbonate reacts with hydrochloric acid.

[1 mark]

Question 3 continues on the next page

Turn over ►

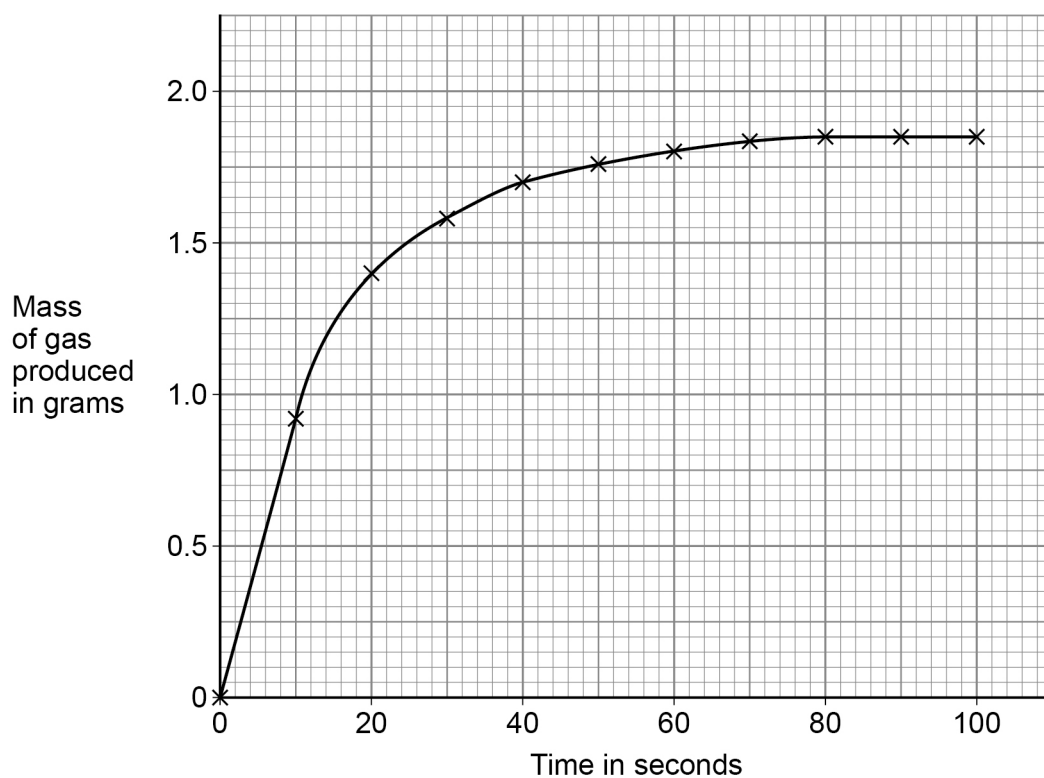


The student determined the mass of gas produced every 10 seconds for 100 seconds.

Do not write
outside the
box

Figure 4 shows the results.

Figure 4



0 3 . 3

Give **three** conclusions from the results in **Figure 4**.

[3 marks]

- 1 _____
- 2 _____
- 3 _____



0 3 . 4 Determine the mean rate of reaction for the first 20 seconds.

Use **Figure 4**.

[3 marks]

Mean rate of reaction = _____ g/s

0 3 . 5 The student repeats the experiment using a higher concentration of hydrochloric acid.

Explain why changing the concentration of the hydrochloric acid affects the rate of the reaction.

[3 marks]

Question 3 continues on the next page

Turn over ►



A different student investigated the rate of reaction of magnesium carbonate with hydrochloric acid.

Describe a method for the investigation.

Your method should give valid results.

[illegible]

17

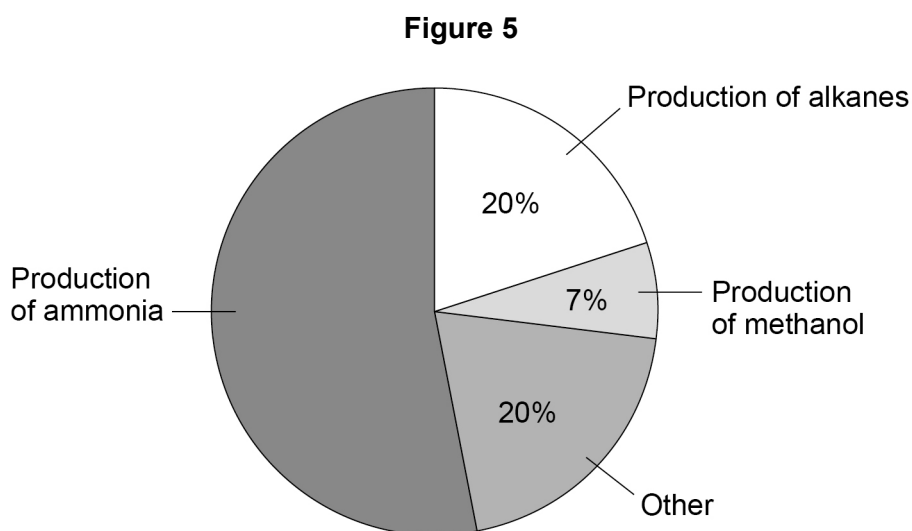


0 4

This question is about ammonia (NH_3).

0 4 . 1

Hydrogen is used to produce ammonia.

Figure 5 shows different uses of hydrogen.

Calculate the percentage (%) of hydrogen used to produce ammonia.

[1 mark]

Percentage of hydrogen = _____ %

0 4 . 2

Ammonia dissolves in water to produce an alkaline solution.

Which ion makes solutions alkaline?

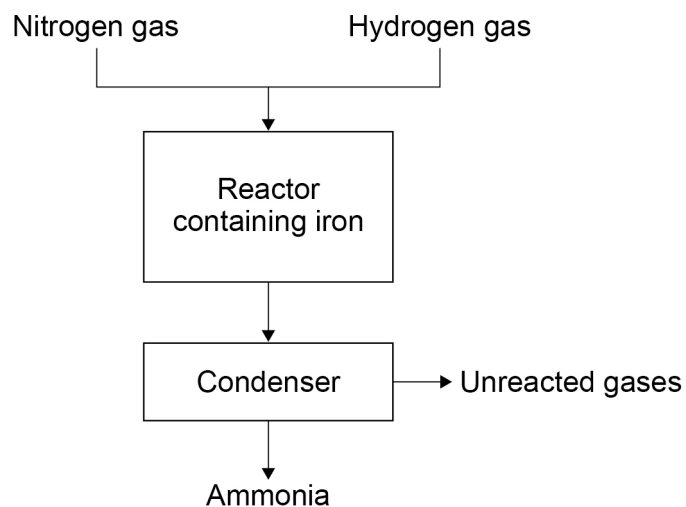
[1 mark]Tick (✓) **one** box.

H ⁺	☐	O ²⁻	☐	OH ⁻	☐
----------------	--------------------------	-----------------	--------------------------	-----------------	--------------------------

Question 4 continues on the next page**Turn over ►**

Figure 6 shows a flow diagram for the production of ammonia in the Haber process.

Figure 6



0 4 . 3

What is the source of the nitrogen used in the Haber process?

[1 mark]

Tick (✓) **one** box.

Air

☐

Crude oil

☐

Plants

☐

Seawater

☐

0 4 . 4

Give an approximate value of the temperature and of the pressure used in the Haber process.

[2 marks]

Temperature _____

Pressure _____



0 4 . 5 The mixture of gases from the reactor is separated in the condenser.

Suggest why ammonia liquefies but the unreacted gases do **not** liquefy in the condenser.

Use **Figure 6**.

[1 mark]

0 4 . 6 What happens to the mixture of unreacted gases when the gases leave the condenser?

Use **Figure 6**.

[1 mark]

0 4 . 7 Iron is used as the catalyst in the Haber process.

Explain why the use of catalysts in industrial processes reduces costs.

[2 marks]

0 4 . 8 Calculate the total number of **atoms** in 5 moles of ammonia (NH₃) molecules.

One mole of ammonia contains 6.02×10^{23} molecules.

[2 marks]

Total number of atoms = _____

11

Turn over ►



0 5

This question is about alcohols and carboxylic acids.

0 5 . 1

Ethanol is an alcohol.

What is the name of the reaction used to produce ethanol from sugar?

[1 mark]Tick (✓) **one** box.

Combustion

☐

Displacement

☐

Fermentation

☐

Neutralisation

☐**0 5 . 2**Ethene (C_2H_4) reacts with substance **X** to produce ethanol ($\text{C}_2\text{H}_5\text{OH}$).Name substance **X**.**[1 mark]**



Table 2 shows the names and formulae of the first three members of a homologous series of alcohols.

Table 2

Alcohol	Formula
Methanol	CH ₃ OH
Ethanol	CH ₃ CH ₂ OH
Propanol	CH ₃ CH ₂ CH ₂ OH

0 5 . 3

How do the formulae of alcohols show that they are members of the same homologous series?

Use **Table 2**.

[1 mark]

0 5 . 4

Give **one** use of propanol.

[1 mark]

0 5 . 5

Propanol can be oxidised to produce propanoic acid.

Complete **Figure 7** to show the displayed structure of propanoic acid (CH₃CH₂COOH).

[1 mark]

Figure 7



Question 5 continues on the next page

Turn over ►



0 5 . 6

Table 3 shows information about the first three members of a homologous series of carboxylic acids.

Table 3

Carboxylic acid	Formula	pH of 1.0 mol/dm ³ solution
Methanoic acid	HCOOH	2.91
Ethanoic acid	CH ₃ COOH	3.39
Propanoic acid	CH ₃ CH ₂ COOH	3.44

Describe how the pH changes when the number of carbon atoms in the carboxylic acid changes.

[1 mark]

0 5 . 7

Carboxylic acids are weak acids.

What is meant by a '**weak** acid'?

[1 mark]



0 5 . 8

A sample of an organic compound **Y** contains:

- 1.92 g carbon
- 0.32 g hydrogen
- 1.28 g oxygen

Determine the empirical formula **and** the molecular formula of organic compound **Y**.

Relative atomic masses (A_r): C = 12 H = 1 O = 16

Relative formula mass (M_r) of **Y** = 88

[6 marks]

Empirical formula = _____

Molecular formula = _____

13

Turn over for the next question

Turn over ►



There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**



0	6
---	---

Aluminium is extracted from aluminium oxide (Al_2O_3) by electrolysis.

Aluminium oxide contains aluminium ions and oxide ions.

0	6	.	1
---	---	---	---

Explain why aluminium is **not** extracted by reacting aluminium oxide with carbon.

[2 marks]

0	6	.	2
---	---	---	---

Give **one** reason why electrolysis is expensive.

[1 mark]

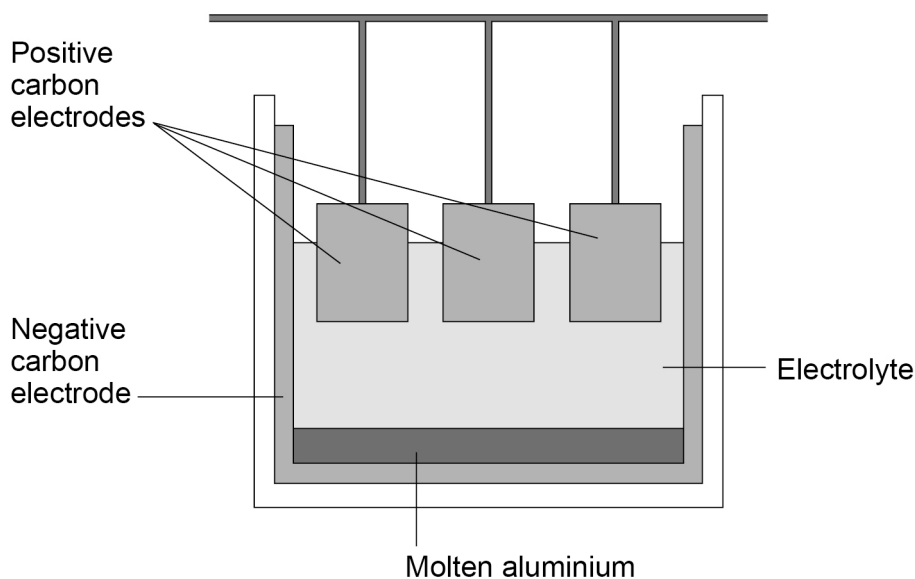
Question 6 continues on the next page

Turn over ►



Figure 8 shows the equipment used for the extraction of aluminium.

Figure 8



0 6 . 3 The electrolyte is a molten mixture of aluminium oxide and **one** other substance.

Name the other substance in the electrolyte.

[1 mark]

0 6 . 4 Why does the molten mixture of aluminium oxide and one other substance conduct electricity?

[1 mark]

0 6 . 5 Aluminium ions are attracted to the negative electrode.

Why is the reaction at the negative electrode described as **reduction**?

[1 mark]



0 6 . 6 Oxygen is produced at the positive electrodes.

Complete the half equation for the reaction that occurs when oxygen is produced.

[2 marks]



0 6 . 7 Explain why the positive electrodes must be replaced regularly.

[3 marks]

0 6 . 8 Write a **balanced** equation for the production of aluminium and oxygen from aluminium oxide.

[2 marks]



13

Turn over for the next question

Turn over ►

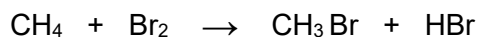


0 7

This question is about reactions of methane.

Methane reacts with bromine when sunlight is present.

The equation for the reaction is:

**0 7 . 1**

The reaction between methane and bromine is exothermic.

What is meant by an 'exothermic reaction'?

[1 mark]

0 7 . 2

Sunlight provides the activation energy for the reaction.

What is meant by 'activation energy'?

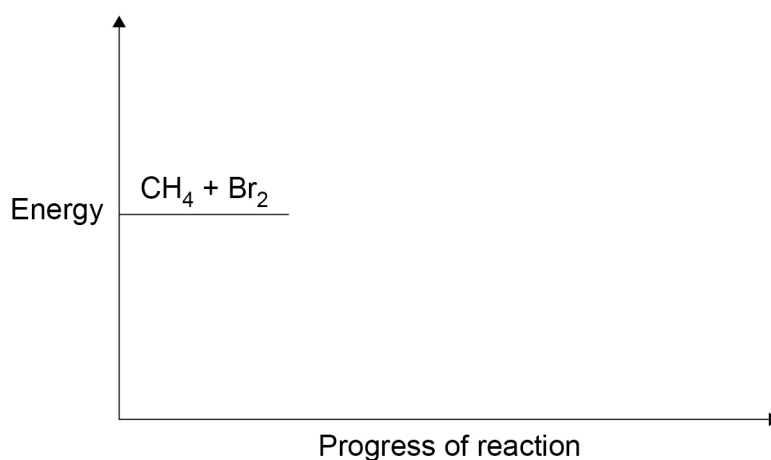
[1 mark]



0 7 . 3

Figure 9 shows part of an energy level diagram for the exothermic reaction between methane and bromine.

Figure 9



Complete **Figure 9**.

You should:

- label the activation energy
- label the overall energy change
- include the formulae of the products.

[4 marks]

Question 7 continues on the next page

Turn over ►



0 7 . 4

Figure 10 shows the displayed structures for the reaction between methane and bromine.

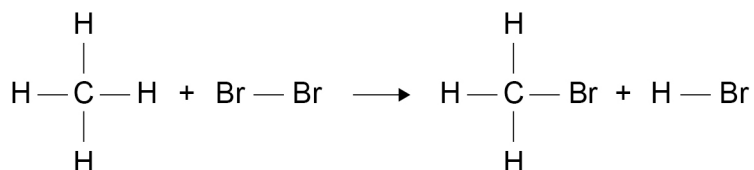
Figure 10

Table 4 shows the bond energies.

Table 4

Bond	C—H	Br—Br	C—Br	H—Br
Bond energy in kJ/mol	412	193	290	366

Calculate the enthalpy change (ΔH) of the reaction.

Use **Figure 10** and **Table 4**.

[4 marks]

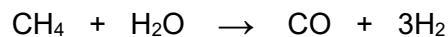
Enthalpy change (ΔH) = _____ kJ/mol



Methane reacts with steam to produce carbon monoxide and hydrogen.

The reaction is endothermic.

The equation for the reaction is:



0 7 . 5 Explain why the reaction is endothermic.

Refer to bonds broken and bonds formed in your answer.

[3 marks]

0 7 . 6 1200 g of methane is completely reacted with steam.

Calculate the maximum volume of hydrogen gas produced.

The volume of one mole of a gas is 24.0 dm^3 .

Relative atomic masses (A_r): C = 12 H = 1

[5 marks]

Maximum volume of hydrogen gas = _____ dm^3

END OF QUESTIONS



There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**



Do not write
outside the
box

[illegible]

*Do not write
outside the
box*

[illegible]

*Do not write
outside the
box*

[illegible]

There are no questions printed on this page

*Do not write
outside the
box*

**DO NOT WRITE ON THIS PAGE
ANSWER IN THE SPACES PROVIDED**

Copyright information

For confidentiality purposes, all acknowledgements of third-party copyright material are published in a separate booklet. This booklet is published after each live examination series and is available for free download from www.oxfordaqaexams.org.uk.

Permission to reproduce all copyright material has been applied for. In some cases, efforts to contact copyright-holders may have been unsuccessful and Oxford International AQA Examinations will be happy to rectify any omissions of acknowledgements. If you have any queries please contact the Copyright Team.

Copyright © 2022 Oxford International AQA Examinations and its licensors. All rights reserved.



3 2



2 2 B Y 9 2 0 2 / 1

IB/M/Nov22/9202/1